Chapter 8, Lecture 1

Marten Wegkamp

EXAMPLE. Assume we have collected lifetimes $X_1, \ldots, X_n$ of n similar transistors. A transistor has the property that it doesn't wear out, or its lifetime has *no memory* in that

$$\mathbb{P}(X > t + \delta | X > t) = \mathbb{P}(X > \delta).$$

One can prove that the only continuous distribution with this property is the Exponential distribution. Assuming that each lifetime is an independent realization, we have an iid sample from a density

$$f(x|\lambda) = \lambda \exp(-\lambda x), x \geq 0.$$

Problem: Find λ.
This will be a common setting. Given iid observations, estimate the parameter.

An ad-hoc solution is as follows. The celebrated *law of large numbers* (Theorem A at page 178 in your textbook) says that the average $\overline{X}_n = \sum_{i=1}^n X_i/n$ converges in probability to $\mathbb{E}(X)$, as $n \to \infty$, meaning,

$$\lim_{n\to\infty} \mathbb{P}\{|\overline{X}_n - \mathbb{E}(X)| > \varepsilon\} = 0 \qquad \forall \varepsilon > 0.$$

Now recall, for an Exponential distribution with parameter λ,

$$\mathbb{E}(X) = \int_0^\infty x(\lambda \exp(-\lambda x))\, \mathrm{d}x = 1/\lambda.$$

Hence,

$$\lim_{n\to\infty} \mathbb{P}\left\{\left|\overline{X}_n - \lambda^{-1}\right| > \varepsilon\right\} = 0 \qquad \forall \varepsilon > 0$$

and

$$\lim_{n\to\infty} \mathbb{P}\left\{\left|(1/\overline{X}_n) - \lambda\right| > \varepsilon\right\} = 0 \qquad \forall \varepsilon > 0.$$

Thus, with high probability, for large n, $1/\overline{X}_n$ is close to λ, the parameter of interest. We *estimate* λ by $\widehat{\lambda} := 1/\overline{X}_n$.

We will see later that this ad-hoc method is part of a general principle, called the method of moments estimation procedure. We will mainly discuss two estimation procedures: the method of moments (MOM) and the method of maximum likelihood (MLE).

The next step is to derive some properties of the estimator. Since it is a random variable, we can compute its expectation, variance, and more generally, its probability distribution (called *sampling* distribution). Let's compute its expected value. In order to do this exact calculation, we need the following facts.

(1) We recall that a Gamma(α, λ) density is of the form

$$f(x|(\alpha,\lambda)) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1}\lambda^{\alpha}\exp(-\lambda x),\ x \geq 0$$

with

$$\Gamma(\alpha) := \int_0^{\infty} x^{\alpha-1}\lambda^{\alpha}\exp(-\lambda x)\,\mathrm{d}x = \int_0^{\infty} y^{\alpha-1}\exp(-y)\,\mathrm{d}y,$$

and

$$\Gamma(\alpha) = (\alpha-1)!,\ \alpha \in \mathbb{N}.$$

See page 53 of the textbook.

(2) Page 98 of the textbook shows that the sum of two independent Exponential(λ) random variables has a Gamma$(2, \lambda)$ distribution. In general, this can be generalized as follows: the sum $X + Y$ of two independent $X \sim \text{Gamma}(\alpha, \lambda)$ and $Y \sim \text{Gamma}(\beta, \lambda)$ has a Gamma$(\alpha + \beta, \lambda)$ distribution.

Now we are all set for our calculation:

$$\begin{aligned}
\mathbb{E}[\widehat{\lambda}] &= \mathbb{E}\left[\frac{1}{\overline{X}_n}\right] \\
&= n\mathbb{E}\left[\frac{1}{T}\right] \qquad \text{with } T = \sum_{i=1}^{n} X_i \\
&= n\int_0^\infty \frac{1}{t}\frac{1}{(n-1)!}\lambda^n t^{n-1}\exp(-\lambda t)\,\mathrm{d}t \qquad \text{since } T \text{ has a Gamma}(n,\lambda) \text{ distribution} \\
&= \frac{n}{n-1}\lambda\int_0^\infty \frac{1}{(n-2)!}\lambda^{n-1}t^{n-2}\exp(-\lambda t)\,\mathrm{d}t \\
&= \frac{n}{n-1}\lambda \qquad \text{Why?}
\end{aligned}$$

We say that $\widehat{\lambda}$ is biased, since

$$\mathbb{E}[\widehat{\lambda}] = \frac{n}{n-1}\lambda \neq \lambda$$

although it is *asymptotically unbiased*, as

$$\lim_{n\to\infty} \mathbb{E}[\widehat{\lambda}] = \lambda.$$

It will be an exercise to derive the *variance* of our estimator:

$$\mathrm{Var}(\widehat{\lambda}) = \frac{n^2}{(n-1)^2}\frac{\lambda^2}{n-2}.$$

Related to these two properties, is the *mean squared error* of our estimator (MSE). It measures how much the estimator deviates from its target (in squared norm) on

average:

$$\begin{aligned}
\mathrm{MSE}(\widehat{\lambda}) &= \mathbb{E}[(\widehat{\lambda}-\lambda)^2] \\
&= \mathbb{E}\left[\{(\widehat{\lambda}-\mathbb{E}[\widehat{\lambda}])+(\mathbb{E}[\widehat{\lambda}]-\lambda)\}^2\right] \\
&= \mathbb{E}\left[(\widehat{\lambda}-\mathbb{E}[\widehat{\lambda}])^2\right]+\mathbb{E}\left[(\mathbb{E}[\widehat{\lambda}]-\lambda)^2\right]+2\mathbb{E}\left[(\widehat{\lambda}-\mathbb{E}[\widehat{\lambda}])(\mathbb{E}[\widehat{\lambda}]-\lambda)\right] \\
&= \mathbb{E}\left[(\widehat{\lambda}-\mathbb{E}[\widehat{\lambda}])^2\right]+\mathbb{E}\left[(\mathbb{E}[\widehat{\lambda}]-\lambda)^2\right] \\
&= \mathrm{Var}(\widehat{\lambda})+\mathrm{Bias}^2(\widehat{\lambda}) \\
&= \frac{1}{(n-1)^2}\lambda^2+\frac{n^2}{(n-1)^2}\frac{\lambda^2}{n-2}.
\end{aligned}$$

We see that the MSE is of order $O(1/n)$, as $n \to \infty$. We will see that this is usually the case.

The next concept is *consistency*. We continue with our example. Let

$$g(t) = 1/t, \qquad t > 0$$

This is a continuous function, for all $t \neq 0$. We observe that

$$\widehat{\lambda} = \frac{1}{\overline{X}_n} = g(\overline{X}_n)$$

and

$$\lambda = g(\mu) \text{ with } \mu := \mathbb{E}[X_1] = 1/\lambda.$$

Suppose $\mu := \mathbb{E}[X_1] \neq 0$. (The case $\mu = 0$ is not very interesting, as it implies that $\mathbb{P}(X = 0) = 1$.) Continuity of $g(t)$ at $\mu \neq 0$ means that for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|\overline{X} - \mu| < \delta$ implies $|g(\overline{X}) - g(\mu)| < \varepsilon$. Hence, we get

$$\begin{aligned}
\mathbb{P}\{|\widehat{\lambda}-\lambda| < \varepsilon\} &= \mathbb{P}\{|g(\overline{X}_n)-g(\mu)| < \varepsilon\} \\
&\geq \mathbb{P}\{|\overline{X}_n-\mu| < \delta\} \\
&\to 1 \qquad \text{by the law of large numbers (LLN)}
\end{aligned}$$

We have argued that our estimator $\widehat{\lambda}$ is *consistent*; it consistently estimates λ and we write

$$\widehat{\lambda} \xrightarrow{P} \lambda, \quad \text{as} n \to \infty.$$

This is a short way of writing that for all $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\{|\widehat{\lambda} - \lambda| < \varepsilon\} = 1,$$

or, taking complements,

$$\lim_{n\to\infty} \mathbb{P}\{|\widehat{\lambda} - \lambda| \geq \varepsilon\} = 0.$$

We used the LLN and a continuity argument, which we will later formalize as the *continuous mapping theorem*.

We will use some simulations in R to illustrate these results.

In the next lecture, we will look at the approximate sampling distribution. The exact sampling distribution of $\widehat{\lambda}$ is n divided by a Gamma(n, λ), a fact we used in our calculation of its expected value before. Using the *δ-method*, we will see that this distribution is close to a normal $N(\lambda, \lambda^2/n)$ distribution, at least for n large.